

Initial Model Training Code, Validation and Evaluation

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Project Name: Diabetic Retinopathy Classification Web Application

1. Introduction

This phase focuses on the development and training of a Convolutional Neural Network (CNN) model for the classification of Diabetic Retinopathy using retinal fundus images.

The model is trained to classify images into the following categories:

- No DR
 - Mild
 - Moderate
 - Severe
 - Proliferative DR
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2. Model Architecture

The model is developed using TensorFlow and Keras.

Architecture includes:

- Convolutional Layers
- MaxPooling Layers
- Flatten Layer
- Dense Fully Connected Layers
- Softmax Output Layer

The final output layer consists of 5 neurons representing the five severity stages.

3. Training Configuration

- Optimizer: Adam
 - Loss Function: Categorical Crossentropy
 - Metrics: Accuracy
 - Epochs: 10–20
 - Batch Size: 32
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4. Model Validation

The dataset is divided into:

- Training Set (80%)
- Validation Set (20%)

Validation accuracy is monitored during training to prevent overfitting.

5. Model Evaluation

After training:

- Accuracy is calculated
- Loss curves are analyzed
- Confusion matrix is evaluated

The model achieves satisfactory accuracy suitable for deployment in the web application.

6. Model Saving

The trained model is saved as:

dr_model.h5

This model is later integrated into the Flask web application for real-time predictions.